

10/9/23

MATHS

YOUVA

Date:

$$36 = 2^2 \times 3^2$$

$$45 = 3^2 \times 5$$

1. $\text{LCM}(36, 45) = 2^2 \times 3^2 \times 5 = 180 \text{ sec} = 3 \text{ min}$

$$36 = 2^2 \times 3^2$$

$$45 = 3^2 \times 5$$

(3) 3

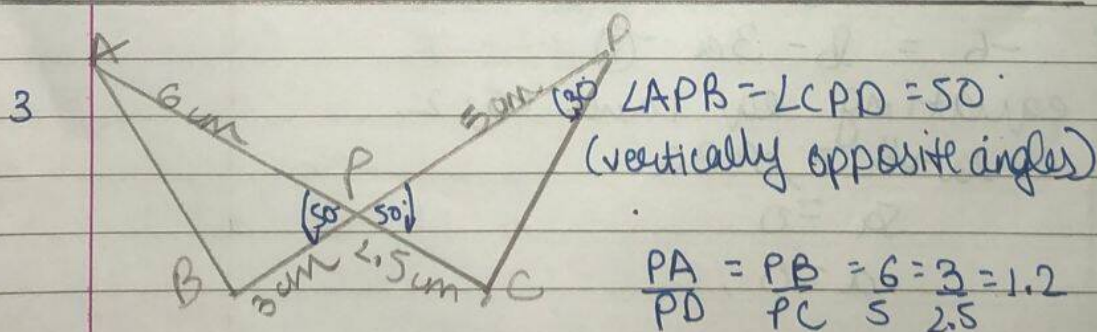
2. $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} \Rightarrow \frac{2 \left(\frac{1}{\sqrt{3}} \right)}{1 - \left(\frac{1}{\sqrt{3}} \right)^2} = \frac{\frac{2}{\sqrt{3}}}{\frac{3-1}{3}}$

$$= \frac{2/\sqrt{3}}{2/3}$$

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{3}{\sqrt{3}} = \sqrt{3}$$

(3) $\tan 60^\circ$ 

by SAS similarity $\triangle APB \sim \triangle DPC$
 $\angle PAB = \angle PDC = 30^\circ$

$$\angle PBA = 180 - (50 + 30) = 180 - 80 = 100^\circ \quad (4) 100^\circ$$

4

$$\triangle ABC \sim \triangle QRP$$

$$\frac{ac(ABC)}{ac(PQR)} = \frac{9}{4} = \frac{BC^2}{PR^2}$$

$$\frac{9}{4} = \frac{(15)^2}{(PR)^2}$$

$$\frac{9}{4} = \frac{225}{PR^2}$$

$$PR^2 = 100$$

$$PR = 10 \text{ cm}$$

(1) 10cm

5

$$p(x) = x^2 + (a+1)x + b$$

$$p(2) = (2)^2 + (a+1)(2) + b$$

$$0 = 4 + 2a + 2 + b$$

$$0 = 6 + 2a + b$$

$$(-6) = 2a + b \quad \text{--- (1)}$$

$$p(-3) = (-3)^2 + (a+1)(-3) + b$$

$$0 = 9 - 3a - 3 + b$$

$$-6 = b - 3a \quad \text{--- (2)}$$

equating eq 1 and eq 2

$$2a + b = b - 3a$$

$$5a = 0$$

$$a = 0$$

6

$$\frac{14587}{2 \times 5^4}$$

$$2 \times 5^4$$

$$\text{denominator} = 2^m \times 5^n$$

$$2 \times 5^4 = 2^m \times 5^4$$

$$m = 1$$

$$n = 4 \quad \because m < n$$

the no. will terminate after 4 decimal places

(4) four decimal places

5

$$x = a$$

$$y = b$$

$$x - y = 2$$

$$a - b = 2 \quad \text{--- (1)}$$

$$x + y = 4$$

$$a + b = 4 \quad \text{--- (2)}$$

add eq 1 and eq 2

$$a - b + a + b = 2 + 4$$

$$2a = 6$$

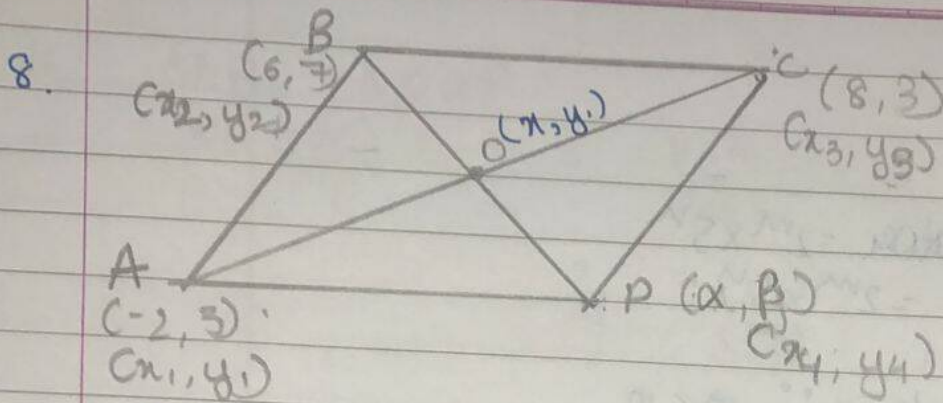
$$a = 3$$

Put a in eq 1

$$3 - 2 = b$$

$$1 = b$$

(3) 3 and 1



Join AC and BD and intersect at O.

∵ diagonals bisect each other in a parallelogram (O is mid point of both the diagonals).

$$x = \frac{x_2 + x_4}{2} = \frac{x_1 + x_3}{2}$$

$$\left[\frac{6 + x_4}{2} = \right]$$

$$\frac{6 + x}{2} = \frac{(-2) + 8}{2}$$

$$6 + x = 6$$

$$x = 0$$

$$y = \frac{y_2 + y_4}{2} = \frac{y_3 + y_1}{2}$$

$$\frac{7 + \beta}{2} = \frac{3 + 3}{2}$$

$$7 + \beta = 6$$

$$\beta = -1$$

$$\boxed{(0, -1)}$$

$$\boxed{\text{Ans: } (0, -1)}$$

9.

$$9x + 3y + 12 = 0$$

$$18x + 6y + 26 = 0$$

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & \\ a_2 & b_2 & c_2 & \end{array} \right] \frac{1}{2} \frac{9}{18} = \frac{3}{6} \neq \frac{12}{26}$$

$$\frac{1}{2} \neq \frac{1}{2} \neq \frac{6}{13}$$

(4) no solution

10

$$2x - 3y + 10 = 0$$

$$3x + ky + 15 = 0$$

as coincident lines

$$\frac{2}{3} = \frac{(-3)}{k} = \frac{10}{15}$$

$$\frac{2}{3} = \frac{(-3)}{k}$$

$$k = \left(\frac{-9}{2} \right)$$

(1) $k = \left(\frac{-9}{2} \right)$

11

$$10x - 5y + 15 = 0$$

$$6x + ky + 19 = 0$$

as intersecting lines

$$\frac{5}{3} \neq \frac{(-5)}{k}$$

$$k \neq -3$$

(1) $k \neq (-3)$

12 $HCF(6, 8, 32) = 2^3 = 8$
 $616 = 2^3 \times 7 \times 11$
 $32 = 2^5$

$\therefore$ 8 rows
(1) 8

13 (1) similar

14 $a = x^3 y^4 z^2$
 $b = x^2 y^4 z^4$
 $LCM(a, b) = x^3 y^4 z^4$

(1) $x^3 y^4 z^4$

15 $\alpha = (-3)$
 $\beta = (4)$
 $\alpha + \beta = -3 + 4 = 1$
 $\alpha\beta = (-3) \times 4 = (-12)$
 $k [x^2 - x(\alpha + \beta) + \alpha\beta]$
 $k [x^2 - x - 12]$
 $k \left[\frac{x^2}{2} - \frac{x}{2} - 6 \right]$

ans: (3) $\frac{x^2}{2} - \frac{x}{2} - 6$

$$\begin{array}{r} 2 \overline{) 616} \\ 2 \overline{) 308} \\ 2 \overline{) 154} \\ 7 \overline{) 77} \\ 11 \overline{) 11} \\ 1 \end{array}$$

17 (1) parallel

18 ~~Let~~ $\Delta ABC \sim \Delta PQR$
 $\frac{ar(ABC)}{ar(PQR)} = \frac{AB^2}{PQ^2} = \frac{(4)^2}{(9)^2} = \frac{16}{81}$

(4) 16:81

19 A: $x^2 - 4x - 5$
 $x^2 - 5x + x - 5 = 0$
 $x(x-5) + (x-5) = 0$
 $(x+1)(x-5) = 0$
 $x = 5; (-1)$

R: $\alpha = 2 + \sqrt{3}$
 $\beta = 2 - \sqrt{3}$
 $\alpha + \beta = 4$
 $\alpha\beta = (4-3) = 1$
 $p(x) = x^2 - 4x + 1$

(3) Assertion (A) is true but Reason R is false.

16 (3) intersecting or coincident

20 $3(R-1)x + 4y - 24 = 0$
 $15x + 20y - 8(R+13) = 0$

$$\frac{3(R-1)}{18} = \frac{4}{20} = \frac{1}{5}$$

$$R-1 = 1$$

$$R = 2$$

(1) Both A and R are true and R is correct explanation of A.

To prove:

$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is a composite no.

$$725 = 5 \times 5 \times 29 \times 1$$

$$\begin{array}{r|l} 5 & 725 \\ 5 & 145 \\ 29 & 29 \\ & 1 \end{array}$$

$\therefore$ it has more than two factors the number is composite number.
 Hence Proved.

22

$$4x + 6y = 7 \Rightarrow 4x + 6y - 7 = 0$$

$$Rx + 18y = 21 \Rightarrow Rx + 18y - 21 = 0$$

$$a_1 = 4$$

$$b_1 = 6$$

$$c_1 = (-7)$$

$$a_2 = R$$

$$b_2 = 18$$

$$c_2 = (-21)$$

As a system of linear equation of infinite solutions.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{4}{R} = \frac{6}{18}$$

$$\frac{4}{R} = \frac{1}{3}$$

$$R = 12$$

$\therefore R = 12$

23

let distance between origin and $(-6, 8)$ be AB

$$x_1 = 0 \quad y_1 = 0$$

$$x_2 = (-6) \quad y_2 = 8$$

Applying distance formula.

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = AB$$

$$\sqrt{0 + (-6)^2 + (0 - 8)^2} = AB$$

$$\sqrt{36 + 64} = AB$$

$$\sqrt{100} = AB$$

$$AB = 10u.$$

$\therefore$ distance between origin and $(-6, 8)$ is 10u.

24.

$$\sin 3A = \cos(A - 10)$$

$$\sin A = \cos(90 - A)$$

$$\text{so } \sin 3A = \cos(90 - 3A)$$

$$\cos(90 - 3A) = \cos(A - 10)$$

on comparing angles.

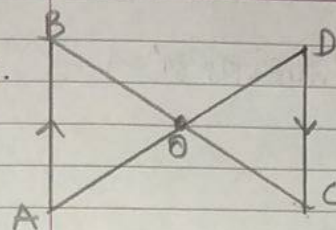
$$90 - 3A = A - 10$$

$$100 = 4A$$

$$25 = A$$

$\therefore$ A is equal to 25.

25.



given $AB \parallel DC$.

to prove $\triangle AOB \sim \triangle DOC$

proof.

$AB \parallel DC$.

$\angle ABO = \angle DCO$ [alternate interior angles]

$\angle BAO = \angle CDO$ [alternate interior angles]

by AA similarity criterion

$$\triangle AOB \sim \triangle DOC$$

Hence proved.

Topic: To prove:

26. $\sin(50^\circ + \theta) - \cos(40^\circ - \theta) + \tan 1^\circ \cdot \tan 10^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ = 1$

proof $\sin(50^\circ + \theta) = \cos(90^\circ - (50^\circ + \theta))$
 $= \cos(40^\circ - \theta)$

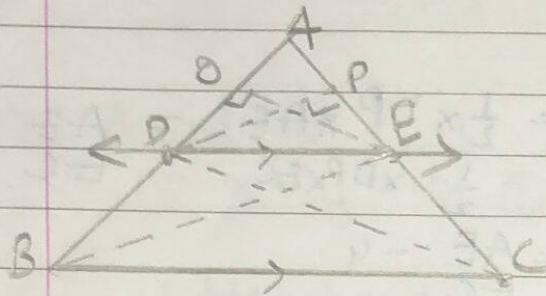
$\tan 1^\circ = \cot(90^\circ - 1^\circ)$
 $= \cot 89^\circ$

$\tan 10^\circ = \cot(90^\circ - 10^\circ)$
 $= \cot 80^\circ$

$\sin(50^\circ + \theta) - \cos(40^\circ - \theta) + \tan 1^\circ \cdot \tan 10^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ$
 $\cos(40^\circ - \theta) - \cos(40^\circ - \theta) + \cot 89^\circ \cdot \tan 89^\circ \cdot \cot 80^\circ \cdot \tan 80^\circ$
 $\frac{0 + (1)(1)}{0 + 1} \left[\because \tan \theta \times \cot \theta = \tan \theta \times \frac{1}{\tan \theta} = 1 \right]$
 1

Hence proved.

27 Basic proportionality theorem states that, if a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points then other two sides are divided in same ratio.



given $ED \parallel BC$
 to prove $\frac{AD}{BD} = \frac{AE}{EC}$

construct (1) Join BE and CD.

(2) then draw a perpendicular EO which is ~~perp~~ $\perp$ at 90° or such that $AD \perp EO$.

(3) draw a perpendicular DP such that $DP \perp AE$

area of $\triangle ADE = \frac{1}{2} \times AD \times EO$ ①

area of $\triangle BDE = \frac{1}{2} \times BD \times EO$ ②

dividing eq 1 by eq 2

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EO}{\frac{1}{2} \times BD \times EO} = \frac{AD}{BD} \quad \text{③}$$

similarly, $\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle CDE)} = \frac{\frac{1}{2} \times DP \times AE}{\frac{1}{2} \times DP \times EC} = \frac{AE}{EC}$

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle CDE)} = \frac{AE}{EC} \quad \text{④}$$

area $[\triangle BDE] = \text{area}[\triangle CDE]$ ⑤ $\left[\begin{array}{l} \because \text{they lie on same} \\ \text{base DE and between} \\ \text{11}^{\text{th}} \text{ lines} \end{array} \right]$

from eq 4 and 5

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle BDE)} = \frac{AE}{EC} \quad \text{⑥}$$

from eq 3 and eq 6

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Hence proved.

28

let,
fixed charge be ~~supers~~ 'x' rupees
extra charge be ~~supers~~ 'y' rupees.
(per day)

case I: Latika paid ₹ 22.
 $x + 4y = 22$ ①

case II: Anand paid ₹ 16.
 $x + 2y = 16$ ②

subtract eq 2 from eq 1

$$\begin{aligned} [x + 4y] - [x + 2y] &= 22 - 16 \\ x + 4y - x - 2y &= 6 \\ 2y &= 6 \\ y &= ₹ 3 \end{aligned}$$

substitute $y = 3$ in eq 1

$$\begin{aligned} x + 4(3) &= 22 \\ x &= 22 - 12 \\ x &= 10 \end{aligned}$$

$\therefore$ fixed charges is ₹ 10/- and extra charge per day is ₹ 3/-

area of quadrilateral = area of $\triangle ABC$ + area of $\triangle ADC$

$$= \frac{35}{2} + \frac{109}{2}$$

$$= \frac{35+109}{2}$$

$$= \frac{144}{2}$$

$$= 72u^2$$

$\therefore$ area of quadrilateral is $72u^2$

30. 1353 and 861.
HCF(1353, 861) =

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

$$123 = 123 \times 1 + 0$$

=

30. HCF(1353, 861) =

$$1353 = 861 \times 1 + 492$$

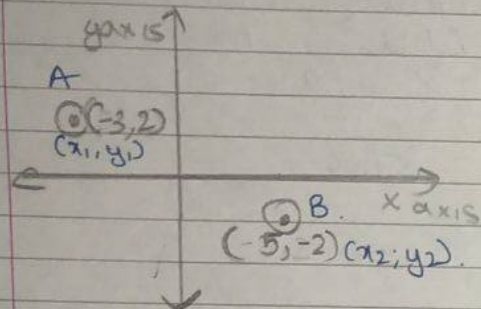
$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

$\therefore$ HCF of 1352 and 861 is 123

31



$$A(-3, 2)$$

$$B(5, -2)$$

$$D(x, 0)$$

$$x_1 = -3$$

$$y_1 = 2$$

$$x_2 = 5$$

$$y_2 = -2$$

$$x_3 = x$$

$$y_3 = 0$$

$$AD = BD$$

Applying distance formula

$$\sqrt{(x_1 - x_3)^2 + (y_1 - y_3)^2} = \sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2}$$

Squaring on both the sides

$$(x_1 - x_3)^2 + (y_1 - y_3)^2 = (x_2 - x_3)^2 + (y_2 - y_3)^2$$

$$(-3 - x)^2 + (2 - 0)^2 = (5 - x)^2 + (-2)^2$$

$$\left[\begin{aligned} \because (a-b)^2 &= a^2 + b^2 - 2ab \\ (a+b)^2 &= a^2 + b^2 + 2ab \end{aligned} \right]$$

$$9 + x^2 - 6x + 4 = 25 + x^2 - 10x + 4$$

$$x^2 - 6x = 20$$

$$x^2 - 6x - 20 = 0$$

$$x^2 - 10x + 4x - 20 = 0$$

$$x(x-10) + 4(x-10) = 0$$

$$(x-10)(x+4) = 0$$

$$x-10=0 \quad | \quad x+4=0$$

$$x=10 \quad | \quad x=-4$$

$\therefore$ points on x axis are 10 and -4

32

$$\frac{\tan \theta + \cot \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta}$$

$$\frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}} \left[\because \tan \theta = \frac{\sin \theta}{\cos \theta} \quad \cot \theta = \frac{\cos \theta}{\sin \theta} \right]$$

$$\frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)}$$

$$\frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} - \frac{\cos^2 \theta}{\sin \theta (\sin \theta - \cos \theta)}$$

$$\frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$\frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{(\sin \theta \cos \theta)(\sin \theta - \cos \theta)} \left[a^3 - b^3 = (a-b)(a^2 + b^2 + ab) \right]$$

$$\frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} \quad [\sin^2 \theta + \cos^2 \theta = 1]$$

$$1 + \sec \theta \csc \theta$$

Hence proved.

33.

$$x - y + 1 = 0$$

$$\text{if } x = 2$$

$$2 - y + 1 = 0$$

$$3 = y$$

$x - y + 1 = 0$			
x	0	1	2
y	1	2	3

$$3x + 2y - 12 = 0$$

$$\text{if } x = 2$$

$$6 + 2y - 12 = 0$$

$$y = 3$$

$$\text{if } x = 0$$

$$0 + 2y - 12 = 0$$

$$y = 6$$

$$\text{if } x = 0$$

$$0 - y + 1 = 0$$

$$1 = y$$

$$\text{if } x = 1$$

$$1 - y + 1 = 0$$

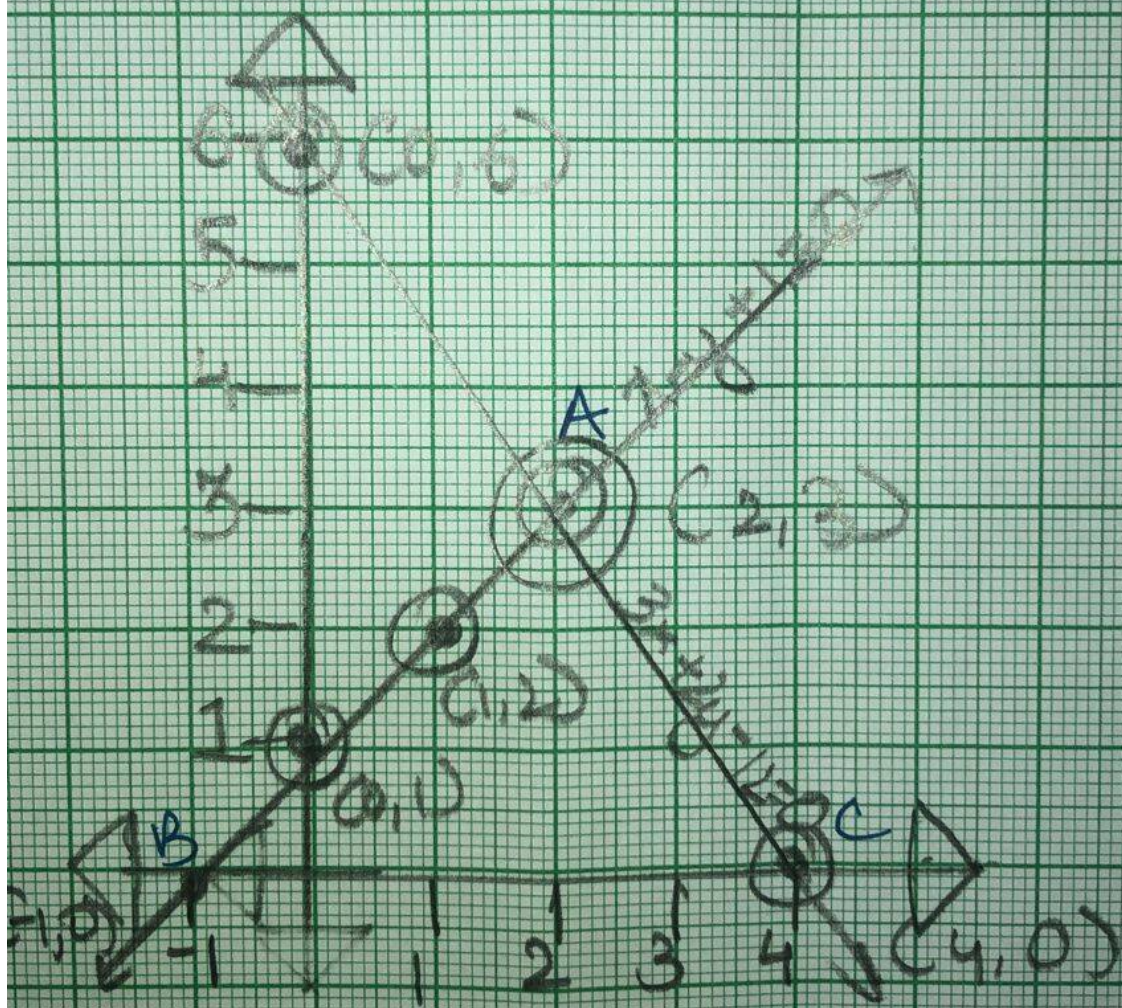
$$2 = y$$

$3x + 2y - 12 = 0$			
x	0	2	4
y	6	3	0

$$\text{if } x = 0$$

$$0 + 2y - 12 = 0$$

$$y = 6$$



vertices of ΔABC .

A - (2, 3)

B - (-1, 0)

C - (4, 0)

$x_1 = 2$

$y_1 = 3$

$x_2 = -1$

$y_2 = 0$

$x_3 = 4$

$y_3 = 0$

area of $\Delta ABC = \frac{1}{2} [x_1(y_2 - y_3)] + [x_2(y_3 - y_1)] + [x_3(y_1 - y_2)]$

$= \frac{1}{2} [2(0 - 0)] + [-1(0 - 3)] + [4(3 - 0)]$

$= \frac{1}{2} [0] + [3] + [12]$

$= \frac{15}{2} u^2$

$= 7.5 u^2$

$\therefore$ area of Δ $7.5 u^2$

34. $x = \sqrt{2}$ $x = -\sqrt{2}$
 $x - \sqrt{2} = 0$ $x + \sqrt{2} = 0$

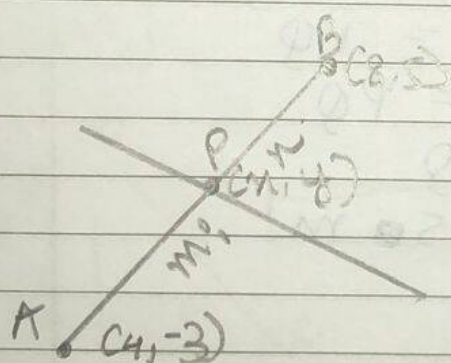
$$(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2 \quad [\because a^2 - b^2 = (a - b)(a + b)]$$

$$\begin{array}{r}
 2x - 1 \\
 x^2 - 2 \overline{) 2x^3 - 4x - x^2 + 2} \\
 \underline{2x^3 - 4x} \\
 0 - x^2 + 2 \\
 \underline{-x^2 + 2} \\
 0
 \end{array}$$

So $2x - 1 = 0$
 $x = \frac{1}{2}$

all zeroes of polynomial $2x^3 - 4x - x^2 + 2$ are $\sqrt{2}$, $-\sqrt{2}$ and $\frac{1}{2}$

35.



$m \circ n = 3:1$
 $m = 3n$
 $n = x$

$x_1 = 4$ $x_2 = 8$
 $y_1 = -3$ $y_2 = 5$

$$\begin{aligned}
 x &= mx_2 + nx_1 \\
 &\quad m+n \\
 &= \frac{3x(8) + x(4)}{3x+x} \\
 &= \frac{4x(6+1)}{4x} \Rightarrow \boxed{x=7}
 \end{aligned}$$

$$\begin{aligned}
 y &= my_2 + ny_1 \\
 &\quad m+n \\
 &= \frac{3x(5) + x(-3)}{3x+x} \\
 &= \frac{3x(5-1)}{4x} \Rightarrow \boxed{3=y}
 \end{aligned}$$

coordinates of point P is $(7, 3)$.

36. (i) B $(0, 12)$

Ans: (2) $(0, 12)$

(ii) P $(3, 5)$

Ans: (1) $(3, 5)$

(iii) Q $(6, 9)$ ~~Let dist~~

P $(3, 5)$

$$x_1 = 3 \quad y_1 = 5$$

$$x_2 = 6 \quad y_2 = 9$$

Applying distance formula.

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = PQ$$

$$\sqrt{(3 - 6)^2 + (5 - 9)^2} = PQ$$

$$\sqrt{(-3)^2 + (-4)^2} = PQ$$

$$\sqrt{9 + 16} = PQ$$

$$\sqrt{25} = PQ$$

$$PQ = 5 \text{ m}$$

(3) 5m.

37 (i) (3) parabola

(ii) (3) 0.

(iii) $(-3, 2)$

Ans: (3) $(-3, 2)$

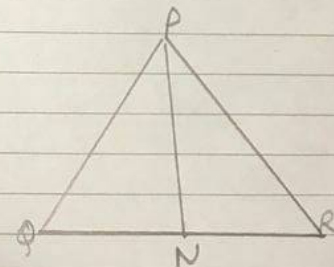
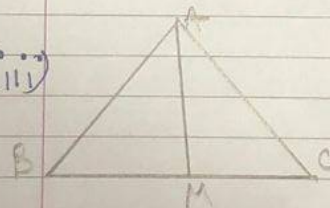
38 Area of kite = $2 \times (\text{area of } \triangle ABC)$

$$= 2 \left(\frac{1}{2} \times 6 \times 8 \right)$$

$$= 48 \text{ cm}^2$$

(i) (1) RHS

(ii)



$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{AM^2}{PN^2}$$

$$\sqrt{\frac{4}{9}} = \frac{AM}{PN}$$

$$2:3 = AM:PN$$

Ans: (1) 2:3